IPL SEASON PERFORMANCE OVERVIEW (2008–2024)

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INTRODUCTION

The Indian Premier League (IPL) is more than just a cricket tournament; it generates a large amount of data from every ball, match, and season. I chose this domain because it represents a complex and dynamic data environment, similar to financial markets or retail cycles, where past patterns and external factors like venue and toss can strongly influence match outcomes. I developed an end-to-end Business Intelligence Dashboard using 16 years of IPL data (2008-2024) . This project treats the IPL not as a game, but as a business case study in performance and strategy. I designed the dashboard in two layers: an Executive View to track organizational KPIs like Output (Runs) and Efficiency (Win %), and a Strategy View that uses conditional probability to quantify the impact of variables like the Toss and Batting Order.

tch_id	inning	batting_team	bowling_team	over	ball	batter	bowler	non_striker	batsman_runs	extra_runs	total_runs	extras_type	is_wicket	player_dismissed	dismissal_kind	fielc
35982	1	Kolkata Knight	Royal Challenge	0	1	SC Gan	P Kumar	BB McCullur	0	1	1	legbyes	0	NA	NA	NA
35982	1	Kolkata Knight	Royal Challenge	0	2	BB Mc	P Kumar	SC Ganguly	0	0	0		0	NA	NA	NA
35982	1	Kolkata Knight	Royal Challenge	0	3	BB Mc	P Kumar	SC Ganguly	0	1	1	wides	0	NA	NA	NA
35982	1	Kolkata Knight	Royal Challenge	0	4	BB Mc	P Kumar	SC Ganguly	0	0	0		0	NA	NA	NA
35982	1	Kolkata Knight	Royal Challenge	0	5	BB Mc	P Kumar	SC Ganguly	0	0	0		0	NA	NA	NA
35982	1	Kolkata Knight	Royal Challenge	0	6	BB Mc	P Kumar	SC Ganguly	0	0	0		0	NA	NA	NA
35982	1	Kolkata Knight	Royal Challenge	0	7	BB Mc	P Kumar	SC Ganguly	0	1	1	legbyes	0	NA	NA	NA

id	season	city	date	match_type	player_of_match	venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
335982	2007/08	Banga	18-04-2008	League	BB McCullum	M Chir	Royal Cl	Kolkata	Royal Challer	field	Kolkata	runs	140	223	20	N	NA	Asad Rauf	RE Koertzer
335983	2007/08	Chanc	19-04-2008	League	MEK Hussey	Punjab	Kings XI	Chenna	Chennai Supe	bat	Chenna	runs	33	241	20	N	NA	MR Benson	SL Shastri
335984	2007/08	Delhi	19-04-2008	League	MF Maharoof	Feroz S	Delhi Da	Rajasth	Rajasthan Ro	bat	Delhi D	wickets	9	130	20	N	NA	Aleem Dar	GA Pratapk
335985	2007/08	Mum	20-04-2008	League	MV Boucher	Wankh	Mumbai	Royal C	Mumbai Indi	bat	Royal C	wickets	5	166	20	N	NA	SJ Davis	DJ Harper
335986	2007/08	Kolka	20-04-2008	League	DJ Hussey	Eden G	Kolkata	Deccan	Deccan Charg	bat	Kolkata	wickets	5	111	20	N	NA	BF Bowden	K Hariharar
335987	2007/08	Jaipu	21-04-2008	League	SR Watson	Sawai I	Rajastha	Kings X	Kings XI Punj	bat	Rajasth	wickets	6	167	20	N	NA	Aleem Dar	RB Tiffin

DATASET

- Dataset Name: IPL Complete Dataset (2008-2024)
- Source: Kaggle
- Time Period: 16 Years (2008 to 2024).
- Nature of Data: Structured Relational Data (CSV format).
- Volume:
  Matches: ~1,095 unique match records.
  Deliveries: ~260,000+ ball-by-ball transaction records.
- Overall, the dataset provides a good combination of time-based, categorical, and numerical features

OBJECTIVES

- To analyze Indian Premier League (IPL) match data and identify key factors influencing match outcomes using interactive data visualizations
- To provide an executive-level overview of IPL performance trends and their evolution over time.
- To evaluate the impact of toss outcomes on match results, assessing whether winning the toss provides a measurable competitive advantage.
- To analyze batting strategy effectiveness, comparing win percentages for teams batting first versus chasing.
- To examine the presence and extent of home advantage by comparing home, away, and neutral venue win rates.

DATA PREPROCESSING

Cleaned and standardized raw IPL data to handle inconsistencies caused by team rebranding, season format changes, and venue name variations.

- Delete Unnecessary column
- Team Name Normalization:(Rising Pune Supergiant & Rising Pune Supergiants, Royal Challengers Bangalore &Royal Challengers Bengaluru,..)
- Season Format Standardization:(2007/08, 2009/10, 2020/21,2009,..)
- Venue De-duplication
- Table Join: Combined match-level and ball-by-ball data using Match ID to create a unified analytical dataset.

MEASURES & CALCULATED FIELDS

Primary Measures

- Total Matches → COUNTD(Match ID)
- Total Runs → SUM(Total Runs)
- Average Runs per Match → SUM(Total Runs) / COUNTD(Match ID)

Calculated Fields

- Season (Cleaned):
  Formula: SPLIT([Season], '/', 1)
- Team Name
- Toss Correlation:
  Formula: IF [Toss Winner] = [Winner] THEN 'YES' ELSE 'NO' END
- Runner-Up Team:
  Formula: IF [Winner] = [Team1] THEN [Team2] ELSE [Team1] END

OVERVIEW & KPI

KPIs Used

- Total Matches
- Total Runshows the total
- Average Runs per Match
- Leading Run Scorer
- Leading Wicket Taker.
- Champion Team
- Runner-up Team

Charts Used

- Line Chart: Season-wise Average Runs per Match
- Bar Chart: Home vs Away Win%, Impact of Toss on Match Result, Batting Strategy Impact on Win %
- Horizontal Bar Chart : Total Runs by Team, Team Win % - Completed Matches

CONCLUSION

- Evolution of the IPL Scoring Environment(bowling Era-Batting Era)
- Average runs per match have increased by over 20%, with a sharp rise in recent seasons.
- High run-scoring teams do not always achieve higher win percentages (RCB, PBKS).
- Winning the toss does not guarantee match success.
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- Historically, chasing provided a slight advantage.
- Recent seasons show a balanced success rate between batting first and chasing.

THANK YOU